

PROBABILITY THEORY

MTM2592-Gr:2

FIFTH WEEK

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SOME IMPORTANT INEQUALITIES

1) Markov Inequality

When the probability function or probability density function of the X random variable is unknown, the first inequality developed to calculate the probabilities related to the variable is the Markov inequality.

Theorem: If X is a random variable that takes non-negative values and c is a constant number; then

$$P(X \geq c) \leq \frac{E(X)}{c}$$

This is Markov Inequality.

Ex: X is the random variable that shows the age of children selected randomly in a children's theater. Since the average of this random variable is $E(X) = 5.5$, calculate the probability that a randomly selected child will be at least 17 by using the Markov inequality.

Solution:

$$P(X \geq 17) \leq \frac{5,5}{17} = 0,32$$

This result reveals the weakness of Markov's inequality. Because those who go to a children's theater must be very low at 17 years old and older.

The Chebyshev inequality derived from the Markov inequality can also be used for all random variables. But Markov's inequality only uses the expected value of a random variable, while Chebyshev's inequality also uses the variance. Thus, more information about the random variable is used.

2) Chebyshev Inequality

THEOREM 8. (*Chebyshev's Theorem*) If μ and σ are the mean and the standard deviation of a random variable X , then for any positive constant k the probability is *at least* $1 - \frac{1}{k^2}$ that X will take on a value within k standard deviations of the mean; symbolically,

$$P(|x - \mu| < k\sigma) \geq 1 - \frac{1}{k^2}, \quad \sigma \neq 0$$

$$P(|x - \mu| \geq k\sigma) \leq \frac{1}{k^2}$$

$$P(|x - \mu| \geq k) \leq \frac{\sigma^2}{k^2}$$

Proof :

- Since $(x - \mu)^2$ is a non-negative random variable, let $c=k^2$. If Markov's inequality is applied to this

$$P\left\{(x - \mu)^2 \geq k^2\right\} \leq \frac{E\left[(x - \mu)^2\right]}{k^2}$$

But $(x - \mu)^2 \geq k^2$

and it is true only for $|x - \mu| \geq k$

$$P\left\{|x - \mu| \geq k\right\} \leq \frac{E\left[(x - \mu)^2\right]}{k^2} = \frac{\sigma^2}{k^2}$$

$$P(X \geq c) \leq \frac{E(X)}{c}$$



Markov's inequality

Note: The importance of Markov's and Cheybshev's inequalities is that they enable us to derive bounds on probabilities when only the mean, or both the mean and the variance, of the probability distribution are known. Of course, if the actual distribution were known, then the desired probabilities could be exactly computed and we would not need bounds.

Ex: Suppose that it is known that the number of items produced in a factory during a week is a random variable with mean 50.

- What can be said about the probability that this week's production will be at least 75?
- If the variance of a week's production is known to equal 25, then what can be said about the probability that this week's production will be between 40 and 60?

Solution:

a) By Markov's inequality

$$P(X \geq c) \leq \frac{E(X)}{c}$$

$$P(X \geq 75) \leq \frac{E(X)}{75} = \frac{50}{75} = \frac{2}{3}$$

b) By Cheybshev's inequality

$$P(|x - \mu| < k\sigma) \geq 1 - \frac{1}{k^2}$$

$$P(40 < X < 60) = P(40 - 50 < X - 50 < 60 - 50) = P(|X - 50| < 10)$$

$$\sigma^2 = 25 \Rightarrow \sigma = 5, \quad k\sigma = 10 \Rightarrow k \cdot 5 = 10 \Rightarrow k = 2$$

$$P(|X - 50| < 10) \geq 1 - \frac{1}{2^2} = 1 - \frac{1}{4} = \frac{3}{4} = 0,75$$

EXAMPLE

If the probability density of X is given by

$$f(x) = \begin{cases} 630x^4(1-x)^4 & \text{for } 0 < x < 1 \\ 0 & \text{elsewhere} \end{cases}$$

find the probability that it will take on a value within two standard deviations of the mean and compare this probability with the lower bound provided by Chebyshev's theorem.

Solution

Straightforward integration shows that $\mu = \frac{1}{2}$ and $\sigma^2 = \frac{1}{44}$, so that $\sigma = \sqrt{1/44}$ or approximately 0.15. Thus, the probability that X will take on a value within two standard deviations of the mean is the probability that it will take on a value between 0.20 and 0.80, that is,

$$\begin{aligned} P(0.20 < X < 0.80) &= \int_{0.20}^{0.80} 630x^4(1-x)^4 dx \\ &= 0.96 \end{aligned}$$

Observe that the statement “the probability is 0.96” is a much stronger statement than “the probability is at least 0.75,” which is provided by Chebyshev's theorem.

$$P(|x - \mu| < k\sigma) \geq 1 - \frac{1}{k^2} \Rightarrow P\left(\left|x - \frac{1}{2}\right| < 2\sigma\right) \geq 1 - \frac{1}{2^2} = \frac{3}{4} = 0.75$$

Example: In a class, students have IQ scores with the mean 70 and standard deviation 4; At least, what percentage of the students have IQ scores between 58 and 82?

Solution: By using Chebyshev's theorem

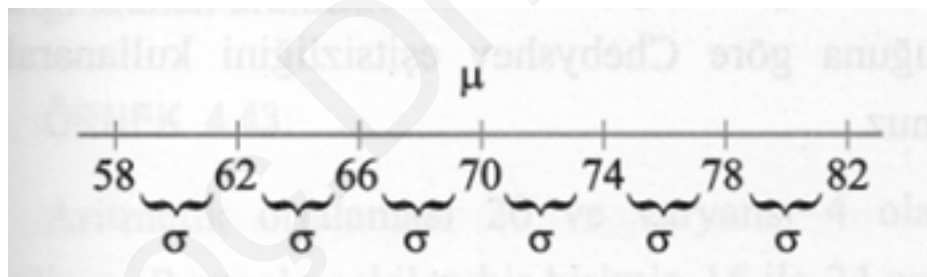
$$P(|x - \mu| < k\sigma) \geq 1 - \frac{1}{k^2}$$

$$P(58 < x < 82) = P(58 - 70 < x - 70 < 82 - 70)$$

$$= P(|x - 70| < 12)$$

$$k\sigma = 12 \Rightarrow k \cdot (4) = 12$$

$$\Rightarrow k = 3$$



$$k = 3$$



$$1 - \frac{1}{3^2} = 1 - \frac{1}{9} = \frac{8}{9} = 0,89 = \%89$$

Example: TV tube production is carried out in a factory. Assuming mean of the weekly production is 1500

- a) what is the probability that this week's television tube production is at least 2000?
- b) If $\sigma^2=1$, then what is the probability that the production will be between 1400 and 1600 in the same week?

Solution: a) By using Markov inequality;

$$P(X \geq c) \leq \frac{E(X)}{c}$$

$$P(X \geq 2000) \leq \frac{E(X)}{2000} = \frac{1500}{2000} = 0,75$$

b) By using Chebyshev's theorem;

$$P(|x - \mu| < k\sigma) \geq 1 - \frac{1}{k^2}$$

$$\begin{aligned} P(1400 < x < 1600) &= P(1400 - 1500 < x - 1500 < 1600 - 1500) \\ &= P(|x - 1500| < 100) \end{aligned}$$

$$k\sigma = 100 \Rightarrow k \cdot (1) = 100$$



$$k = 100$$

$$P\{|X - 1500| < 100\} \geq 1 - \frac{1}{100^2} = 0,99$$

Example: The mean and variance of the weekly cost of a small shop is 1000 TL and 400 TL, respectively. Determine an interval for the weekly cost whose occurring probability is at least 75%.

Ex: X is the random variable that shows the age of children selected randomly in a children's theater. Since the average of this random variable is 5.5 and standard deviation is 1, calculate the probability that a randomly selected child will be at least 17.

MIXED RANDOM VARIABLES

These are random variables that are neither discrete nor continuous, but are a mixture of both. In particular, a mixed random variable has a continuous part and a discrete part. We will use the Dirac delta function to analyze mixed random variables. Technically speaking, the Dirac delta function is not actually a function. It is what we may call a generalized function. Nevertheless, its definition is intuitive and it simplifies dealing with probability distributions.

Definition: Dirac delta function is defined as

$$\delta(x - x_0) = \begin{cases} \infty, & x = x_0 \\ 0, & x \neq x_0 \end{cases}$$

with the condition

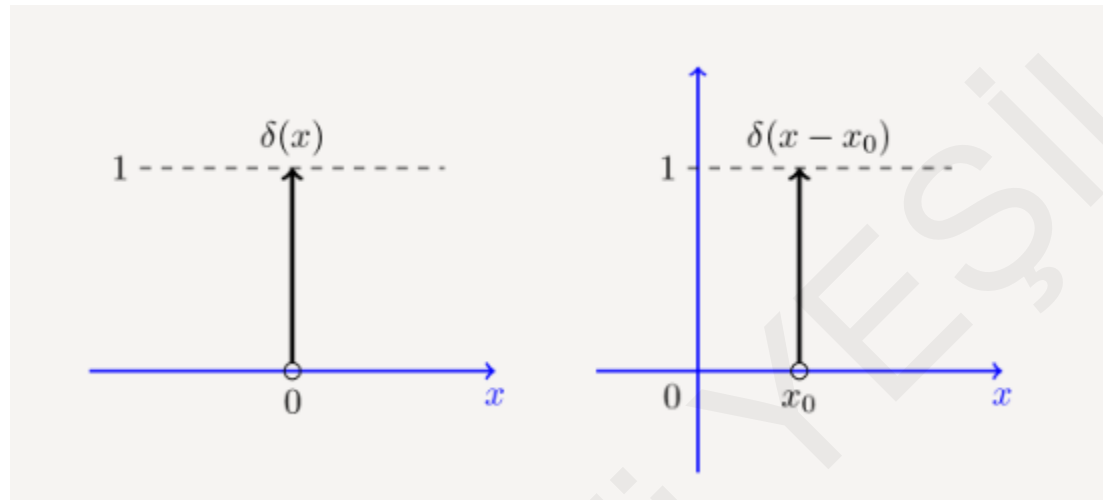
$$\int_{x_0 - \varepsilon_0}^{x_0 + \varepsilon_0} \delta(x - x_0) dx = 1 \quad \text{for any } \varepsilon > 0.$$

$\delta(x - x_0)$ is not a traditional function and no function defined on the real numbers has the properties given above. It can be considered as a probability distribution. It also satisfies;

$$1) \quad \delta(x - x_0) = \frac{d}{dx} u(x - x_0) \quad \text{where } u(x - x_0) = \begin{cases} 1, & \text{if } x \geq x_0 \\ 0, & \text{if } x < x_0 \end{cases}$$

$$2) \quad \int_{x_0 - \varepsilon_0}^{x_0 + \varepsilon_0} g(x) \delta(x - x_0) dx = \int_{-\infty}^{\infty} g(x) \delta(x - x_0) dx = g(x_0)$$

For any $\epsilon > 0$ and any function $g(x)$ continuous over $(x_0 - \epsilon, x_0 + \epsilon)$



Graphical representation of delta function.

Let X be a discrete random variable with range $R_x = \{x_1, x_2, \dots, x_k\}$ and $f(x_i)$, $i=1, 2, \dots, k$ be its probability function. Then, its cumulative function can be written as:

$$F(x) = \sum_{i=1}^k f(x_i) u(x - x_i)$$

which is equivalent to

$$\Rightarrow F(x) = \begin{cases} 0, & x < x_1 \\ f(x_1), & x_1 \leq x \leq x_2 \\ \dots & \\ \sum_{i=1}^t f(x_i), & x_t \leq x \leq x_{t+1} \\ \dots & \\ 1, & x \geq x_k \end{cases}$$

Then, by differentiating it we have

$$f_G(x) = \frac{dF(x)}{dx} = \sum_{i=1}^k f(x_i) \frac{d}{dx} u(x - x_i) = \sum_{i=1}^k f(x_i) \delta(x - x_i)$$

which is called the generalized probability density function of discrete random variable. If $f_G(x)$ does not contain dirac delta, X is continuous. If $f_G(x)$ contains both dirac delta and non-dirac delta functions, X is said to be mixed random variable.

$$f_G(x) = \sum_{i=1}^k f(x_i) \delta(x - x_i) + g(x) \rightarrow \text{Generalized PDF of a mixed random variable}$$

where $g(x) \geq 0$ does not contain dirac delta function and $f_G(x)$ satisfies.

$$1) f_G(x) \geq 0$$

$$2) \int_{-\infty}^{\infty} f_G(x) dx = \underbrace{\sum_{i=1}^k f(x_i) \int_{-\infty}^{\infty} \delta(x - x_i) dx}_{\sum_{i=1}^k f(x_i)} + \int_{-\infty}^{\infty} g(x) dx = 1$$

Accordingly,

$$\mu_x = E(X) = \int_{-\infty}^{\infty} x f_G(x) dx = \sum_{i=1}^k x_i f(x_i) + \int_{-\infty}^{\infty} x g(x) dx$$

and

$$\text{Var}(X) = \int_{-\infty}^{\infty} (x - \mu_x)^2 f_G(x) dx = \sum_{i=1}^k (x_i - \mu_x)^2 f(x_i) + \int_{-\infty}^{\infty} (x - \mu_x)^2 g(x) dx$$

Definition : Properties of the delta function

We define the delta function $\delta(x)$ as an object with the following properties:

1. $\delta(x) = \begin{cases} \infty & x = 0 \\ 0 & \text{otherwise} \end{cases}$
2. $\delta(x) = \frac{d}{dx} u(x)$, where $u(x)$ is the unit step function $u(x) = \begin{cases} 0 & x < 0 \\ 1 & x \geq 0 \end{cases}$);
3. $\int_{-\epsilon}^{\epsilon} \delta(x) dx = 1$, for any $\epsilon > 0$;
4. For any $\epsilon > 0$ and any function $g(x)$ that is continuous over $(x_0 - \epsilon, x_0 + \epsilon)$, we have

$$\int_{-\infty}^{\infty} g(x) \delta(x - x_0) dx = \int_{x_0 - \epsilon}^{x_0 + \epsilon} g(x) \delta(x - x_0) dx = g(x_0).$$

Example:

$$F(x) = \begin{cases} 0, & x < 0 \\ 1/4, & 0 \leq x < 1 \\ \frac{3}{4} + \frac{1}{4}(x-1), & 1 \leq x \leq 2 \\ 1, & x \geq 2 \end{cases}$$

- a) Plot $F(x)$ and find the type of random variable
- b) Find and graph $f_G(x)$
- c) Find $P(X \leq 3/2)$, $P(X > 1)$ and $P(X \geq 1)$
- d) Compute $E(X)$ and $\text{Var}(X)$.